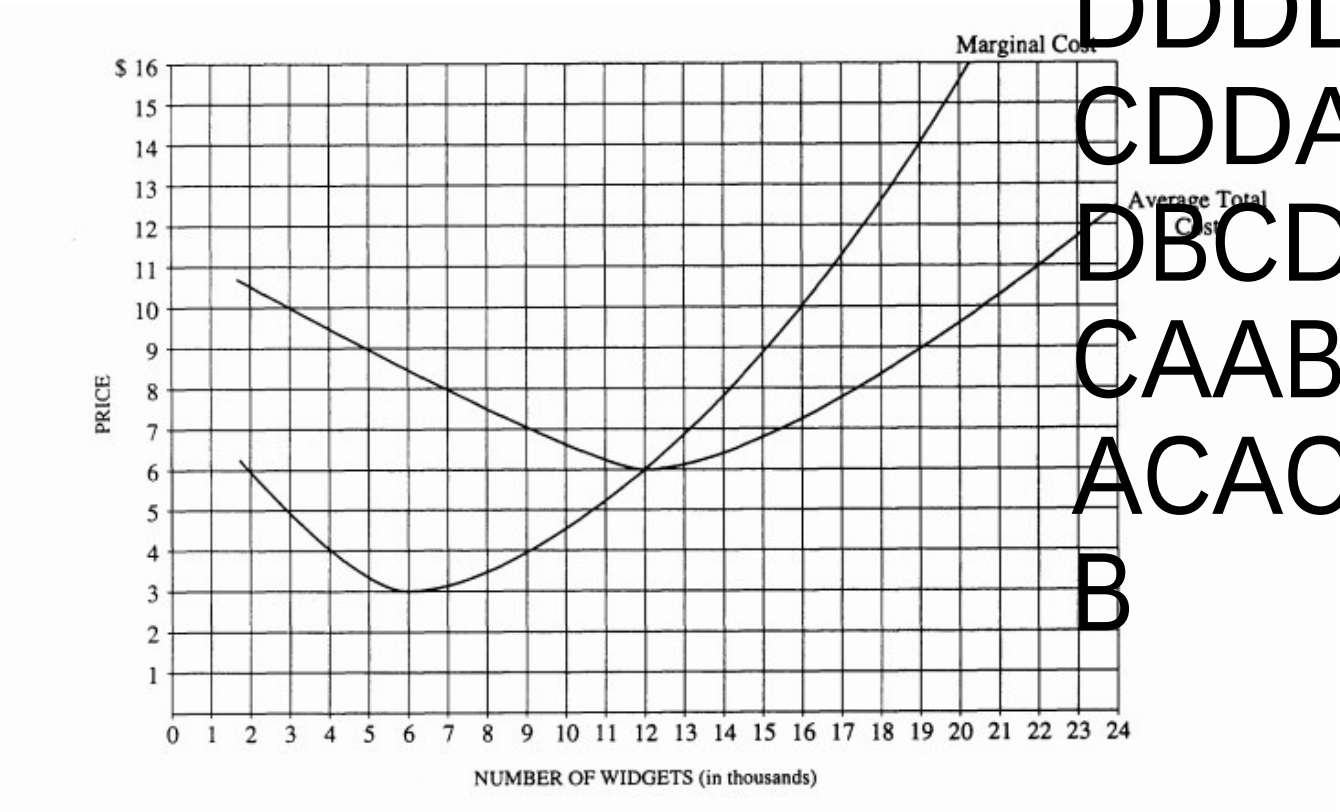
3.7 perfect competition

DCDAE
DDDDC
CDDAB
DBCDE
CAABB
ACACC
B

The following questions refer to the graph below showing cost curves for a perfectly competitive firm.



1. If the market price is $10, how many widgets should this profit-maximizing firm produce?

(A) 3,000
(B) 6,000
(C) 12,000
(D) 16,000 ✓
(E) 21,000

2. At market price $6, the profit-maximizing rate of output will result in

(A) economic profits
(B) economic losses
(C) normal profits ✓
(D) profits that are less than normal
(E) profits that are greater than normal

3.7 perfect competition

3. A constant-cost, perfectly competitive gadget industry is in long-run equilibrium. An increase in the number of consumers of gadgets will most likely result in

(A) higher short-run and long-run prices for gadgets

(B) reduced short-run profits, followed by the exit of some firms

(C) an upward shift in all short-run cost curves, followed by a higher long-run price for gadgets

(D) a higher short-run price for gadgets, followed by an increase in the quantity produced ✓

(E) a decrease in the demand for gadgets, followed by a decrease in the supply of gadgets

4. A constant-cost, perfectly competitive industry is in long-run equilibrium. If the demand for the good increases, which of the following will occur in the long run?

(A) The price will remain unchanged. ✓

(B) The price will increase.

(C) The price will decrease.

(D) Economic profits will increase.

(E) Economic profits will decrease.

5. A farmer produces peppers in a perfectly competitive market. If the price falls, in the short run the farmer should

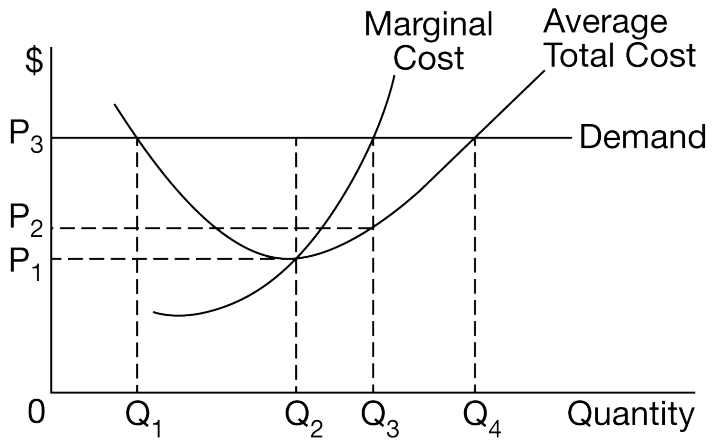
(A) increase production until the new price equals average revenue

(B) increase production to offset the fall in price

(C) discontinue production if the new price is less than marginal revenue

(D) continue to produce only if the new price covers average fixed costs

(E) continue to produce only if the new price covers average variable costs ✓



6. How many units of output should a firm with the cost and demand curves shown above produce to maximize profit?

3.7 perfect competition

(A) 0
(B) Q_1
(C) Q_2
(D) Q_3 ✓
(E) Q_4

Answer D

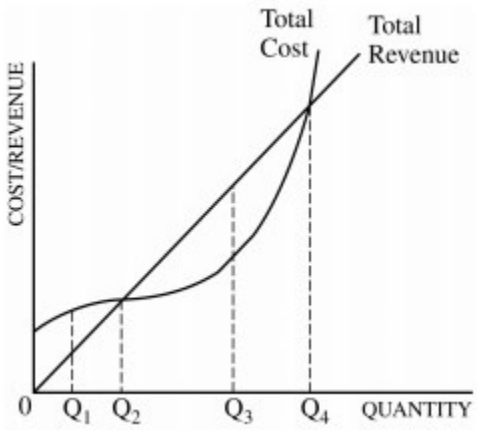
Correct. The profit-maximizing quantity occurs at the output quantity at which marginal revenue equals marginal cost. The firm depicted in the graph operates in a perfectly competitive market; its price is the same as marginal revenue and equal to the horizontal demand curve. To maximize profit the firm should produce where $\mathrm{MR} = \mathrm{MC}$, which occurs at Q_3.

---

7. A firm is producing the allocatively efficient level of output if

(A) total revenue is equal to total cost
(B) marginal revenue is equal to marginal cost
(C) price is equal to average total cost
(D) price is equal to marginal cost ✓
(E) price is equal to total cost

8.



The graph above shows the total revenue and total cost curves for a firm in which type of market structure and what is the profit-maximizing quantity?

3.7 perfect competition

(A)

Market Structure	Quantity
Monopoly	Q_2

(B)

Market Structure	Quantity
Monopoly	Q_3

(C)

Market Structure	Quantity
Perfect Competition	Q_1

(D) ✓

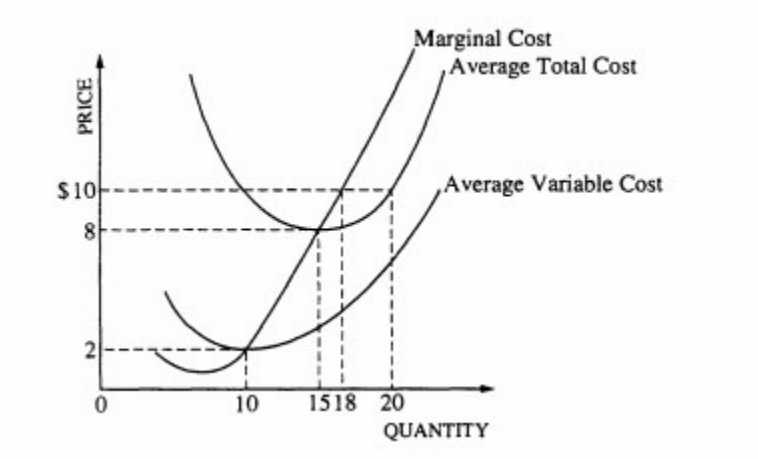
Market Structure	Quantity
Perfect Competition	Q_3

(E)

Market Structure	Quantity
Perfect Competition	Q_4

3.7 perfect competition

9.



The diagram above shows a perfectly competitive firm's short-run cost curves. If the price of the output increases from $8 to $10, the profit-maximizing firm will

(A) continue producing 15 units because average total cost is a minimum

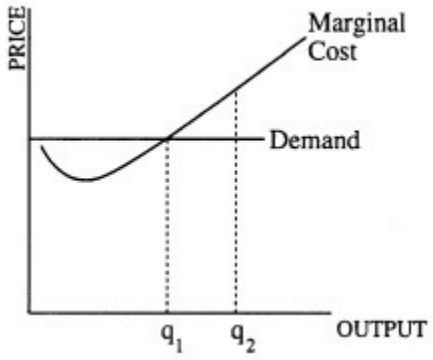
(B) continue producing 15 units because average total cost is equal to marginal cost

(C) increase output to 20 units because this is the output at which price equals average total cost

(D) increase output to 18 units because this is the output at which price equals marginal cost ✓

(E) decrease output to 10 units because this is the output at which average variable cost is at a minimum

10.



Given the cost and demand schedules depicted above, if the firm increased output from q_1 to q_2, it would

(A) earn a normal profit

(B) experience an increase in profits

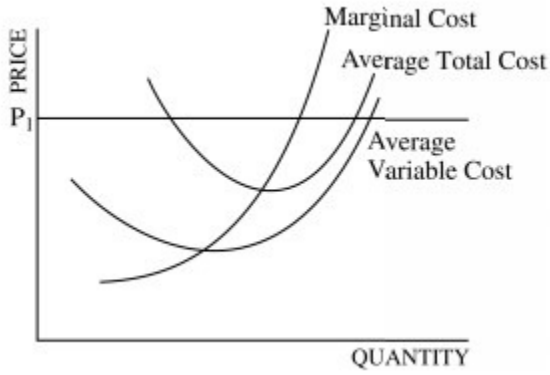
(C) experience a decline in profits ✓

(D) increase revenues but not costs

(E) increase costs but not revenues

3.7 perfect competition

11.



The graph above shows the short-run cost curves of a firm in a perfectly competitive market. Which of the following are true at the firm's profit-maximizing output level?

I. Price exceeds average total cost.

II. Economic profits are zero.

III. Marginal cost equals average total cost.

IV. New firms are likely to enter the market in the long run.

(A) I and II only
(B) I and III only
(C) I and IV only ✓
(D) II, III, and IV only
(E) I, II, III, and IV

12. The allocatively efficient level of output is produced in any market structure when

(A) firms are experiencing economies of scale
(B) marginal revenue is equal to zero
(C) long-run economic profit is equal to zero
(D) price is equal to marginal cost ✓
(E) average variable cost is at a minimum

Answer D

Correct. The allocatively efficient level of output is produced when price is equal to marginal cost.

3.7 perfect competition

13. All of the following are essential characteristics of a perfectly competitive industry EXCEPT:

(A) All products produced by the firms in the industry are homogeneous.

(B) All firms in the industry are price takers.

(C) All consumers and firms are completely aware of the prices at which transactions take place in the industry.

(D) There are barriers to entry into and exit from the industry. ✓

(E) Price is equal to marginal revenue for every firm in the industry.

14. A market is clearly NOT perfectly competitive if which of the following is true in equilibrium?

(A) Price exceeds marginal cost. ✓

(B) Price exceeds average variable cost.

(C) Price exceeds average fixed cost.

(D) Price equals opportunity cost.

(E) Accounting profits are positive.

15. A perfectly competitive firm, earning economic profits, produces and sells 100 units of output at a price of $20 per unit. If its marginal cost of increasing output to a rate of 101 units is $18, which of the following statements is correct?

(A) The total revenue from selling 101 units is the same as the total revenue from selling 100 units

(B) The total profit from selling 101 units is $2 greater than the total profit from selling 100 units. ✓

(C) The total cost of producing 101 units is $2 greater than the total cost of producing 100 units.

(D) To sell 101 units, the firm must reduce its price below $20.

(E) To sell 101 units, the firm must raise its price above $20.

16. A perfectly competitive firm is currently in long-run equilibrium. Its total revenue is $100,000, and the average total cost of production is $100. Which of the following can be concluded from this information?

(A) The firm's marginal cost is $1,000, and its profit is positive.

(B) The firm's marginal cost is $1,000, and its profit is zero.

(C) The firm's output is 1,000 units, and its profit is negative.

(D) The firm's output is 1,000 units, and its profit is zero. ✓

(E) The firm's output is 1,000 units, and its profit is positive.

17. A perfectly competitive firm's short-run supply curve is the portion of the marginal cost curve that is

3.7 perfect competition

(A) above the average total cost curve

(B) above the average variable cost curve ✓

(C) above the minimum average fixed cost curve

(D) above the demand curve

(E) between the average fixed cost curve and the average variable cost curve

18. A perfectly competitive market in equilibrium is allocatively efficient and it maximizes

(A) total consumer surplus

(B) total producer surplus

(C) the sum of total consumer surplus and total producer surplus ✓

(D) total revenue

(E) total external benefits

A profit-maximizing, perfectly competitive firm is currently in long-run equilibrium. It is earning $15,000 of total revenue from a sale of 1,000 units. Its total fixed cost of production is $2,500.

19. Which of the following can correctly be inferred from the information provided?

(A) Its marginal cost is $12.50, and its average total cost is $12.50.

(B) Its marginal cost is $12.50, and its average variable cost is $12.50.

(C) Its marginal cost is $15.00, and its average total cost is $12.50.

(D) Its marginal cost is $15.00, and its average variable cost is $12.50. ✓

(E) Its marginal cost is $15.00, and its average fixed cost is $12.50.

---

20. Assume that a competitive industry producing a normal good is in long-run equilibrium. If average consumer income decreases, which of the following changes will occur?

3.7 perfect competition

(A)

Short-Run Price	Short-Run Industry Output	Movement Firms
Increase	Increase	Enter

(B)

Short-Run Price	Short-Run Industry Output	Movement Firms
Increase	Decrease	Exit

(C)

Short-Run Price	Short-Run Industry Output	Movement Firms
Decrease	Increase	Exit

(D)

Short-Run Price	Short-Run Industry Output	Movement Firms
Decrease	Decrease	Enter

(E) ✓

Short-Run Price	Short-Run Industry Output	Movement Firms
Decrease	Decrease	Exit

21. Assume that a firm that produces a good in a constant-cost perfectly competitive industry is in long-run equilibrium. If the demand for the good increases, the profit-maximizing output by the firm will change in which of the following ways in the short run and long run?

3.7 perfect competition

(A)

Short Run	Long Run
Return to original level	Return to original level

(B)

Short Run	Long Run
Increase	Increase

(C) ✓

Short Run	Long Run
Increase	Return to original level

(D)

Short Run	Long Run
Decrease	Decrease

(E)

Short Run	Long Run
Decrease	Return to original level

22. Assume that a perfectly competitive firm is in long-run equilibrium. If industry demand for the product increases, how will this firm's price, output, and profit change in the short run?

3.7 perfect competition

(A)

Price	Output	Profit
Increase	Increase	Increase

(B)

Price	Output	Profit
Increase	Decrease	Decrease

(C)

Price	Output	Profit
Increase	Increase	Decrease

(D)

Price	Output	Profit
No change	Decrease	Decrease

(E)

Price	Output	Profit
Decrease	Increase	Increase

23. At a perfectly competitive firm's current output level, average total cost is $15, average variable cost is $10, and marginal cost is $8 and increasing. If the product price is $15, what should this firm do to maximize profits?

(A) Increase the quantity of output produced.

(B) Increase the product price.

(C) Decrease the product price to increase sales.

(D) Shut down immediately.

(E) Continue to produce at its current output level.

24. At the current production level of good X, price is greater than marginal cost. Which of the following actions would lead to greater efficiency?

3.7 perfect competition

(A) Decreasing the production of good X

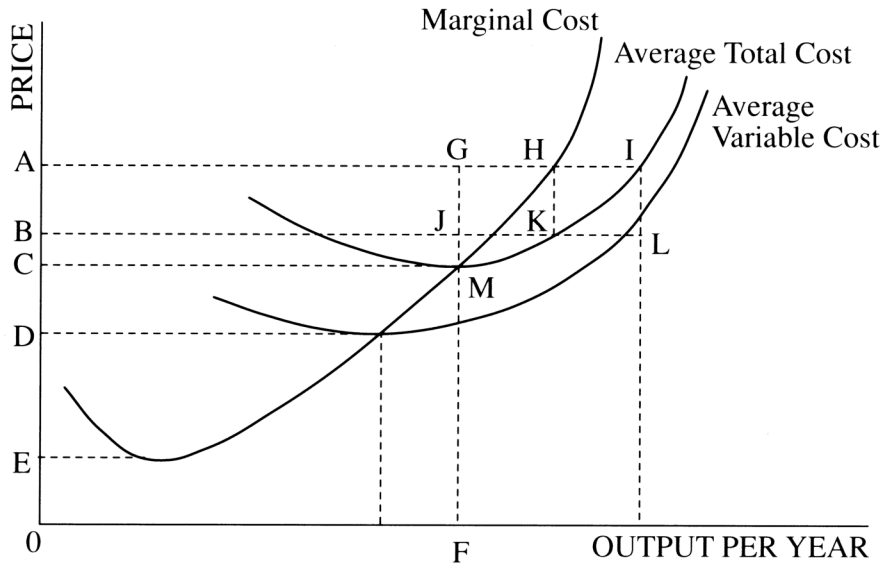
(B) Increasing the production of good X ✓

(C) Maintaining the current level of production of good X

(D) Imposing a tax on the production of good X

(E) Imposing an effective price floor on good X

Refer to the following diagram and assume a perfectly competitive market structure.



25. At the price 0A, economic profits are

(A) ABJG

(B) ABKH ✓

(C) ABLI

(D) ACMG

(E) C0FM

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26. A typical firm in a perfectly competitive constant-cost industry is operating with an economic loss in the short run. When the industry returns to long-run equilibrium, what will happen to the number of firms in the industry, the market price, and the typical firm's quantity?

3.7 perfect competition

(A) ✓

Number of Firms	Market Price	Firm's Quantity
Decrease	Increase	Increase

(B)

Number of Firms	Market Price	Firm's Quantity
Decrease	Decrease	Increase

(C)

Number of Firms	Market Price	Firm's Quantity
Decrease	Increase	Decrease

(D)

Number of Firms	Market Price	Firm's Quantity
Increase	Increase	Decrease

(E)

Number of Firms	Market Price	Firm's Quantity
Increase	Decrease	Decrease

27. Reff Corp is a firm with total revenue of $1,000, marginal cost of $5, and average variable cost of $4. Both the output and the input markets are perfectly competitive, and Reff Corp is currently in long-run equilibrium. Reff Corp's output and total fixed cost of production must be equal to which of the following?

3.7 perfect competition

(A)

Output	Fixed Cost
250	$800

(B)

Output	Fixed Cost
250	$400

(C) ✓

Output	Fixed Cost
200	$200

(D)

Output	Fixed Cost
200	$400

(E)

Output	Fixed Cost
200	$800

Catering, Inc., which provides catering services in a perfectly competitive market, was maximizing profits at the market price of $22 per meal. The market price has recently increased to $28 per meal.

28. Which of the following short-run adjustments will increase profits for Catering, Inc.?

(A) Increasing output until the marginal cost equals the new price ✓
(B) Increasing output until the average total cost equals the new price
(C) Decreasing output because of the increase in market price
(D) Increasing the wage rate paid to its workers
(E) Increasing advertising to increase demand for the meals

29. In the short run in perfect competition, the industry's demand curve and a firm's demand curve have which of the following slopes?

3.7 perfect competition

(A)

Industry's Demand Curve	Firm's Demand Curve
Horizontal	Downward sloping

(B)

Industry's Demand Curve	Firm's Demand Curve
Horizontal	Horizontal

(C) ✓

Industry's Demand Curve	Firm's Demand Curve
Downward sloping	Horizontal

(D)

Industry's Demand Curve	Firm's Demand Curve
Downward sloping	Downward sloping

(E)

Industry's Demand Curve	Firm's Demand Curve
Vertical	Horizontal

30. Assume that all firms in a perfectly competitive market currently earn positive economic profits. What will happen in the long run if all firms face constant returns to scale in production?

(A) The price of the product will increase.

(B) Firms will exit the industry.

(C) The quantity produced by each existing firm will decrease. ✓

(D) Average total cost of production will increase.

(E) Nothing will change.

Answer C

Correct. With constant returns to scale, average costs are at minimum and firms will face the same cost

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curves in the long run. Positive economic profits will cause new firms to enter the market, which shifts the market supply curve to the right, decreasing the market price in the long run. Existing firms are now facing a lower market price, therefore their marginal revenue curve will intersect the marginal cost curve at a lower quantity.

31. Zucchini is produced in a perfectly competitive market with a downward-sloping demand curve and an upward-sloping supply curve. Dawson Farm is a typical perfectly competitive farm that produces and sells zucchini at the equilibrium price of $1.75 per pound. Which of the following is true?

(A) No buyer in the market is willing to pay more than $1.75 per pound for zucchini.

(B) No buyer in the market is willing to pay more than $1.75 per pound for zucchini from Dawson Farm. ✓

(C) Market demand for zucchini will increase if Dawson Farm lowers its price below $1.75 per pound.

(D) Dawson Farm would lose some but not all of its customers if it increases the price above $1.75 per pound for zucchini.

(E) Dawson Farm would likely be able to increase the demand for its zucchini by advertising.

Answer B

Correct. Dawson Farm operates in a perfectly competitive market and therefore is a price taker. No buyer in this market will be willing to pay more than the market price from Dawson Farm, because buyers can buy zucchini from other producers in the market at the same market price of $1.75 per pound.

32. Assume a decreasing-cost perfectly competitive industry. Which of the following statements is true?

(A) Firms will earn economic profits in long-run equilibrium.

(B) The short-run market supply curve is upward sloping; the long-run supply curve is horizontal or perfectly elastic.

(C) As industry output expands, there are fewer firms producing in the long run.

(D) As industry output contracts, each firm's long-run average total cost curve shifts upward. ✓

(E) Input prices rise as the industry produces more output.

Answer D

Correct. With a decreasing-cost industry, as the industry's output expands, the price of at least one input falls as more of that input, like specialized capital, is produced and used. This input price decrease shifts down the long-run average cost of each firm, lowering the minimum of long-run average cost for each

3.7 perfect competition

firm and lowering the long-run output price at which economic profits equal zero. Conversely, when the industry contracts, long-run average total cost shifts upward, and the long-run equilibrium price is higher.

33. Which of the following is true for a perfectly competitive, decreasing-cost industry?

(A) The price of inputs will decrease as the number of firms in the industry decreases.

(B) The long-run market supply curve will be downward sloping. ✓

(C) Firms earn positive economic profit in the long run.

(D) Firms do not have a shutdown condition.

(E) There are no economies of scale.

Answer B

Correct. A perfectly competitive decreasing-cost industry is an industry with a negative sloping long-run industry supply curve that results because expansion of the industry causes lower production cost and resource prices. As the market expands, the old and new firms experience lower costs of production, which makes the new zero economic profit level intersect at a lower price than before.

34. Firm XYZ produces and sells corn in a perfectly competitive market and hires its workers in a perfectly competitive labor market. Which of the following best describes the demand curve for XYZ's corn and XYZ's demand curve for labor?

3.7 perfect competition

(A)

Demand for XYZ's Corn	XYZ's Labor Demand
Horizontal	Horizontal

(B) ✓

Demand for XYZ's Corn	**XYZ's Labor Demand**
Horizontal	Downward sloping

(C)

Demand for XYZ's Corn	XYZ's Labor Demand
Horizontal	Vertical

(D)

Demand for XYZ's Corn	**XYZ's Labor Demand**
Downward sloping	Downward sloping

(E)

Demand for XYZ's Corn	**XYZ's Labor Demand**
Downward sloping	Horizontal

35. Donuts for Dollars is a typical profit-maximizing firm in a perfectly competitive market that is earning positive economic profit in the short run. Which of the following statements must be true?

(A) Price must be less than average total cost in the short run, and firms will exit the market in the long run.

(B) Price must be less than average total cost in the short run, and firms will earn normal profit in the long run.

(C) Price must equal average total cost in the short run, and firms will enter the market in the long run.

(D) Price must be greater than average total cost in the short run, and firms will exit the market in the long run.

(E) Price must be greater than average total cost in the short run, and firms will earn normal profit in the long run. ✓

3.7 perfect competition

Answer E

Correct. Per-unit profit is calculated as price minus average total cost; therefore, if price is greater than average total cost for Donuts for Dollars, they are earning positive economic profit in the short run. Because there are no barriers to entry in perfect competition, short-run economic profits will encourage new firms to enter the market, thus increasing the market supply and decreasing the market price. Donuts for Dollars will earn normal profit (zero economic profit) in the long run, as the market price decreases to equal the firm's average total cost.

36. Which of the following are characteristics of a perfectly competitive industry?

I. New firms can enter the industry easily.

II. There is no product differentiation.

III. The industry's demand curve is perfectly elastic.

IV. The supply curve of an individual firm in the industry is perfectly elastic.

(A) I and II only ✓

(B) I and III only

(C) II and IV only

(D) I, II, and IV only

(E) I, III, and IV only

37. Which of the following is true for a perfectly competitive firm in long-run equilibrium?

(A) It earns positive economic profit.

(B) It is allocatively efficient. ✓

(C) It experiences economic losses.

(D) It is productively inefficient.

(E) It maximizes revenues.

38. If there are many firms in an industry and each firm's product is indistinguishable from the products of all other firms, the individual firm's demand curve will be

(A) upward sloping and different for each firm

(B) downward sloping and different for each firm

(C) downward sloping and identical for every firm

(D) horizontal and different for each firm

(E) horizontal and identical for every firm ✓

3.7 perfect competition

39. Which of the following is true of a firm in a perfectly competitive industry?

(A) It faces a perfectly elastic demand curve. ✓

(B) It faces a downward-sloping demand curve.

(C) It will increase its total revenue if it increases the selling price of its product.

(D) Its cost could be greatly reduced if it did not have to advertise its product.

(E) It sets the price of its product at the level at which profits are maximized or losses are minimized.

40. Suppose that price in a perfectly competitive industry decreases and it is now below minimum average total cost but remains above minimum average variable cost. Which of the following will occur in the short run?

(A) New firms will enter the industry.

(B) Firms will increase output so that marginal revenue equals the new price.

(C) Firms will produce the output at which average total cost is at a minimum.

(D) Firms will produce the output at which marginal cost equals the new price. ✓

(E) Firms will not produce at all, since they will be unable to cover all their costs.

A perfectly competitive firm is producing 10 units of output and sells the product for $\$5$ per unit. At this level of output the average total cost is $\$4$, the average variable cost is $\$3$ and the marginal cost is $\$7$.

41. What should this firm do to maximize short-run profits?

(A) Increase output until price equals average total cost.

(B) Increase output until price equals marginal cost.

(C) Leave output unchanged because price is greater than average total cost.

(D) Decrease output until price is equal to marginal cost. ✓

(E) Decrease output until price is equal to average total cost.

Answer D

Correct. The firm is currently producing where marginal cost ($\$7$) is greater than marginal revenue (which is equal to price, $\$5$). The firm can increase profit by decreasing output to where marginal cost equals price.

---

42. Which of the following is a characteristic of firms in a perfectly competitive industry?

3.7 perfect competition

(A) Engaging in collusive behavior
(B) Facing significant barriers to entry
(C) Setting the market price
(D) Producing identical products ✓
(E) Earning positive long-run economic profit

43. For a perfectly competitive firm producing the profit-maximizing quantity, the average total cost is $10 and the average variable cost is $8. If the market price for its product is $10, which of the following is true for the firm?

(A) It is sustaining a loss and should shut down.
(B) It is earning zero economic profit and will remain in business. ✓
(C) Its accounting profits exceed implicit costs.
(D) It will temporarily shut down until price rises.
(E) It is making an economic profit that will attract other firms to the industry.

44. When a perfectly competitive firm sells additional units of output, its total revenue will

(A) remain constant
(B) increase rapidly at first, then decline
(C) increase at a decreasing rate
(D) increasing at an increasing rate
(E) increasing at a constant rate ✓

45. One characteristic of perfectly competitive markets is that individual firms

(A) engage in product differentiation
(B) are free to enter or exit an industry in the long run ✓
(C) earn positive economic profits in the long run
(D) advertise to increase market share
(E) face a downward-sloping demand curve

46. The reason that firms in perfect competition earn zero economic profit in the long run is that

(A) firms are small
(B) there are a large number of sellers
(C) firms cannot advertise
(D) there are no barriers to entry or exit ✓
(E) the commodities produced are relatively inexpensive

47. Which of the following will most likely lead to zero economic profits?

3.7 perfect competition

(A) Price floors
(B) Price discrimination
(C) External costs
(D) Free entry and exit of firms ✓
(E) External benefits

48. Which of the following situations best fits with the description of a firm in a perfectly competitive market?

(A) GaryNote constantly offers discounts to steal customers from EricNote, which retaliates by offering price cuts.
(B) GaryNote has many competitors in the market, and each firm differentiates its product to gain customer loyalty.
(C) GaryNote can hire as many workers as it wants at the market wage, but it must decrease its price to sell more output.
(D) GaryNote sets its own price for its output, but because of market competition, it earns zero economic profit in the long run.
(E) GaryNote can sell as many units of output as it wants at the market price but will lose all its customers if it charges a higher price. ✓

Answer E

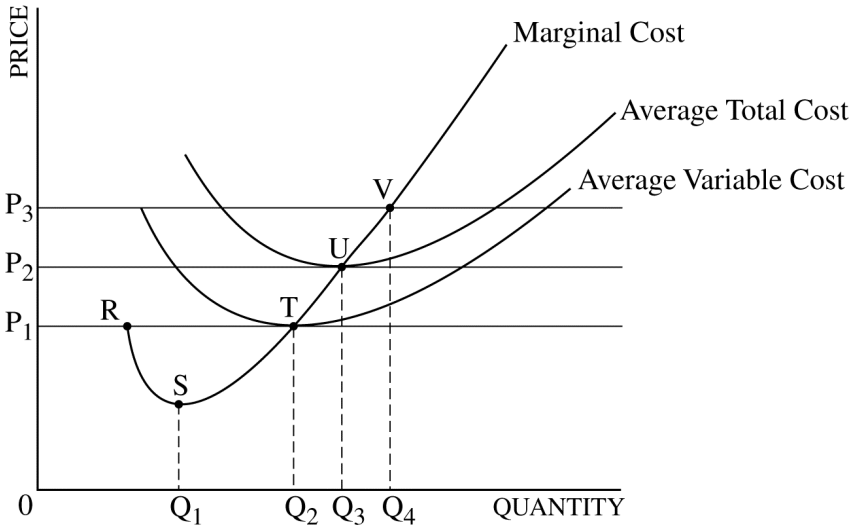
Correct. Since firms in perfectly competitive markets are price-takers, they face a horizontal demand curve for their product at the market price. Therefore, attempting to charge any price above the market price will result in the loss of all of its customers.

49. Which of the following best describes a perfectly competitive market?

(A) Many small firms producing differentiated products and facing significant barriers to entry
(B) Many small firms producing a homogeneous product and facing significant barriers to entry
(C) Many small firms producing a homogeneous product and facing no significant barriers to entry ✓
(D) A single large firm producing a unique product and facing significant barriers to entry
(E) A few large firms producing a differentiated product and facing no significant barriers to entry

3.7 perfect competition

50.



Which of the following segments of the marginal cost curve lies entirely on the firm's short-run supply curve?

(A) TUV ✓

(B) STU

(C) RSTU

(D) RSTUV

(E) RS

51. If a perfectly competitive firm increases its price above the market equilibrium price, which of the following will be true for this firm?

(A) Its total revenue will increase.

(B) Its profit will increase.

(C) Its sales will decrease but profit will not be affected.

(D) Its demand curve will become downward sloping.

(E) It will not be able to sell any output. ✓

52. If a perfectly competitive firm is producing where marginal cost is rising and greater than marginal revenue, to maximize profits it should

(A) increase the level of production

(B) decrease the level of production ✓

(C) maintain current level of production

(D) increase price

(E) decrease price

53. If a perfectly competitive firm wishes to maximize profits and is producing where price exceeds both marginal cost and average variable cost, then the firm is

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(A) producing too little output ✓

(B) producing where its marginal revenue is less than its marginal cost

(C) using too much of the variable input

(D) charging too high a price for its output

(E) not earning any economic profits

54. If a perfectly competitive industry is in long-run equilibrium, which of the following is most likely to be true?

(A) Some firms can be expected to leave the industry.

(B) Individual firms are not operating at the minimum points on their average total cost curves.

(C) Firms are earning a return on investment that is equal to their opportunity costs. ✓

(D) Some factors are not receiving a return equal to their opportunity costs.

(E) Consumers can anticipate price increases.

55. If a perfectly competitive market with no government intervention is allocatively efficient, which of the following must be true?

(A) Consumer surplus plus producer surplus is at its maximum. ✓

(B) Consumer surplus is larger than producer surplus.

(C) Producer surplus is larger than consumer surplus.

(D) All surplus is producer surplus.

(E) All surplus is consumer surplus.

56. If a profit-maximizing firm in a perfectly competitive market chooses to produce in the short run, then marginal cost is always

(A) greater than or equal to total cost

(B) greater than or equal to average total cost

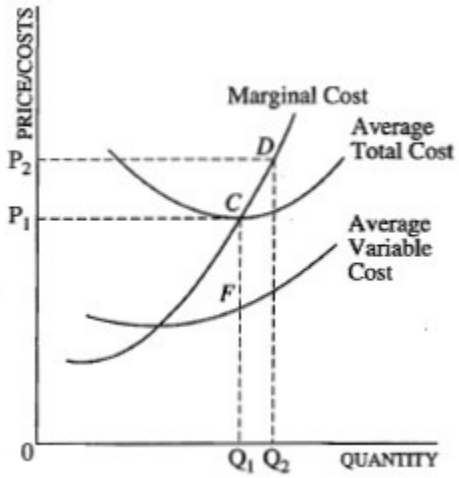
(C) greater than or equal to average variable cost ✓

(D) greater than or equal to average fixed cost

(E) less than average total cost

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The following questions refer to the graph below, which shows the cost curves for a profit maximizing, perfectly competitive firm.



57. If marginal revenue is equal to P_1, all of the following statements are true EXCEPT:

(A) Total revenue will equal total costs.

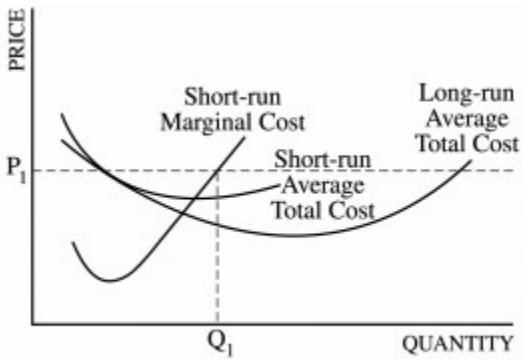
(B) The firm will produce Q_1 units of output.

(C) The firm will produce the efficient level of output.

(D) The firm will earn a normal profit.

(E) The firm will increase production in the long run. ✓

The following questions refer to the graph below, which shows the cost curves of a firm.



58. Which of the following will be true if the firm is in a perfectly competitive market and the price is P_1 ?

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(A) The firm will earn short-run profits but suffer long-run losses.

(B) In the long run, existing firms in the industry will produce an output level greater than Q_1. ✓

(C) In the long run, existing firms will leave the industry.

(D) Firms will leave the industry until profits are increased.

(E) New firms will be dissuaded from entering the industry, at least until the price increases.

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59. In a perfectly competitive industry, the market price of the product is $12. A firm produces at a level of output where average total cost is $16, marginal cost is $16, and average variable cost is $8. To maximize its profit, the firm should

(A) decrease its selling price

(B) increase its selling price

(C) decrease output but keep producing ✓

(D) shut down

(E) level both price and output unchanged

60. Which of the following statements about a constant-cost perfectly competitive industry in long-run equilibrium must be true?

(A) An increase in demand will cause no change in the long-run equilibrium price. ✓

(B) An increase in demand will cause no change in the long-run equilibrium quantity.

(C) The long-run supply curve is upward sloping.

(D) The long-run supply curve is perfectly inelastic.

(E) The total cost of production remains the same as output increases.

61. Assume an increasing-cost perfectly competitive industry. Which of the following statements is true?

(A) The short-run market supply curve will be upward sloping, but the long-run market supply curve will be horizontal.

(B) Firms must be producing where their long-run average cost curve is upward sloping, exhibiting diseconomies of scale.

(C) In the long run, firms will actually sustain economic losses.

(D) Firms no longer produce at the minimum of long-run average total cost in long-run equilibrium.

(E) As the industry expands its output, at least one input price increases, increasing the minimum of long-run average total cost. ✓

Answer E

Correct. With an increasing-cost industry, as the industry's output expands, the increased demand for at

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least one input raises the price of that input. This input price increase shifts up the long-run average cost of each firm, raising the minimum of long-run average total cost for each firm and the long-run output price at which economic profits equal zero.

62. In the short run, a perfectly competitive firm should shut down whenever

(A) minimum average variable cost exceeds price ✓

(B) minimum average total cost exceeds price

(C) marginal revenue exceeds marginal cost

(D) marginal cost exceeds marginal revenue

(E) it is losing money

63. The characteristic that causes firms in a perfectly competitive industry to earn zero economic profits in the long run is

(A) firms are price takers

(B) firms produce identical products

(C) individual firms account for a small fraction of the total market

(D) the industry supply curve is horizontal

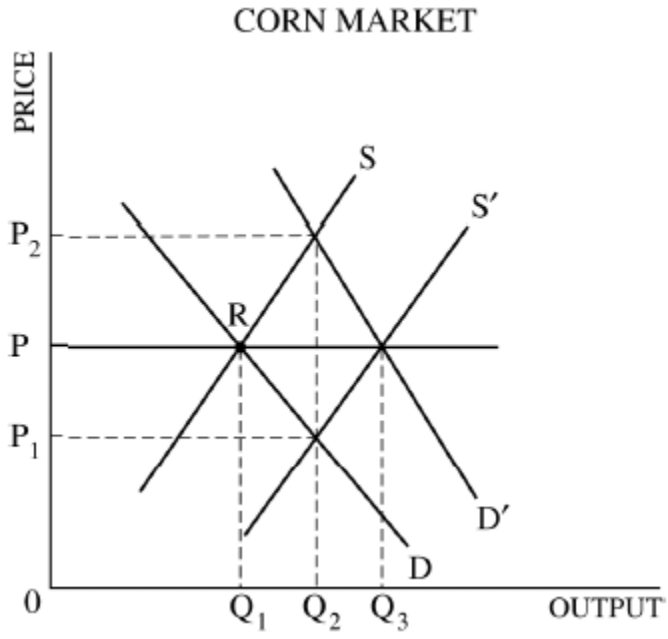
(E) there are no barriers to entry or exit ✓

Answer E

Correct. The key assumption that leads to long-run adjustment in perfectly competitive markets is the assumption of no barriers to entry or exit (easy entry or exit). Depending on the market condition in the short run, firms respond by entering the market if economic profits are positive and exiting the market if economic profits are negative. The entry or exit of firms will stop when each firm earns zero economic profit. That means no firm has an incentive to leave the market or enter the market.

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64. The following question is based on the graph below showing the long-run adjustment of a constant-cost perfectly competitive industry.



The long-run industry supply curve for corn is

(A) upward sloping

(B) downward sloping

(C) horizontal ✓

(D) downward sloping at first and then upward sloping

(E) upward sloping at first and then downward sloping

65. Which of the following is true of a perfectly competitive firm in long-run equilibrium?

(A) It produces its output at minimum average total cost. ✓

(B) It earns positive economic profits.

(C) It will exit the industry.

(D) Its price exceeds marginal cost.

(E) Its price exceeds marginal revenue.

66. Which of the following is true about the marginal revenue of a firm in a perfectly competitive industry?

(A) It is constant. ✓

(B) It increases as output sold increases.

(C) It decreases as output sold increases.

(D) It increases at first, then decreases.

(E) It decreases at first, then increases.

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67. Which of the following is true if a perfectly competitive market is in long-run equilibrium?

(A) Market price will eventually decrease.
(B) New firms will enter the industry.
(C) Marginal revenue is equal to average total cost. ✓
(D) Price is not equal to marginal revenue.
(E) Average variable costs are decreasing.

68. Which of the following market structures results in allocative efficiency?

(A) Monopoly
(B) Monopolistic competition
(C) Perfect competition ✓
(D) Natural monopoly
(E) Oligopoly

69. Assume paper clips are produced in a perfectly competitive industry that is currently in long-run equilibrium. One firm in the industry develops a new production method that reduces its variable cost of production. If its competitors are unable to copy this new method, what will happen to this firm's output and profit in the short run?

(A) Output will increase, and profit will increase. ✓
(B) Output will decrease, and profit will increase.
(C) Output will not change, and profit will increase.
(D) Output will increase, and profit will not change.
(E) Output will not change, and profit will not change.

Answer A

Correct. Firms in perfect competition that can reduce their variable costs will shift their marginal cost, average variable cost, and average total cost curves to the right; this will result in marginal revenue and marginal cost intersecting at an increased quantity of output. Since the firm is a price taker in perfect competition, the combination of an increase in output and a reduction in variable cost must increase their profits.

70. Which of the following statements is true about a firm that sells its output in a perfectly competitive market?

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(A) The demand for its product is a downward-sloping function.

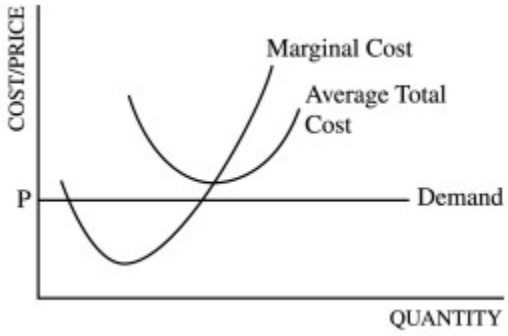
(B) The firm will earn zero economic profits in long-run equilibrium. ✓

(C) Advertising is an important tool of the firm.

(D) The firm will increase its total economic profits if it charges a price that is lower than the market price.

(E) The marginal revenue that firm receives from selling an additional unit of output will be different from the price at which it sells that unit.

The following question refers to the diagram below, which shows the demand and cost curves for a profit-maximizing firm.



71. Which of the following statements best describes the graph?

(A) Economic losses are incurred, and the firm will increase price until no losses are incurred.

(B) Economic losses are incurred, and exit of firms from the market will cause prices to increase in the long run. ✓

(C) Economic profits are earned, and costs will increase until no profits are earned.

(D) Economic profits are earned, and entry of firms into the market will cause prices to decrease in the long run.

(E) Economic profits are earned, and neither exit nor entry of firms will occur in the long run.

---

72. Which of the following is true for a perfectly competitive firm?

(A) The firm is a price taker. ✓

(B) The firm chooses its profit-maximizing quantity by equating price to average total cost.

(C) The firm faces a downward-sloping demand curve for its product.

(D) The firm's short-run supply curve is the same as its average variable cost curve above the minimum of the marginal cost curve.

(E) The firm can expect to earn positive economic profit in long-run equilibrium.

73. Which of the following indicates that a perfectly competitive firm is in long-run equilibrium?

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(A) Price equals marginal cost.
(B) Price equals average revenue.
(C) Price equals marginal cost, which equals average total cost. ✓
(D) Price equals average revenue, which equals marginal revenue.
(E) Price equals average fixed cost.

74. Which of the following statements relating to a profit-maximizing perfectly competitive firm is true?

(A) The firm is facing downward-sloping demand and marginal revenue curves.
(B) The firm will raise its price to increase revenue.
(C) The firm's price is above its marginal revenue at every output level.
(D) The firm's price is given by the market and is equal to marginal revenue. ✓
(E) The firm continues to produce as long as marginal revenue is greater than zero.

Answer D

Correct. The competitive firm is a price taker, accepting the market price as given. As a result, the firm's demand curve is perfectly elastic (or horizontal) at the market price. Its marginal revenue is also horizontal and will equal price at every output level.

75. Which of the following conditions is necessary for productive efficiency?

(A) The firm is producing at the point where the marginal product of labor is greater than the average product of labor.
(B) The firm is producing a given level of output with a least-cost combination of inputs. ✓
(C) Consumer surplus is minimized.
(D) There are external costs.
(E) There is no government regulation.

76. Productive efficiency occurs when a firm produces output at a level at which

(A) average total cost equals average revenue
(B) average total cost is at a minimum ✓
(C) price exceeds average total cost
(D) price equals marginal cost
(E) marginal revenue exceeds average revenue

77. If individual firms in a perfectly competitive market are earning positive economic profits, the number of firms and the price of the product in the market will most likely change in which of the following ways in the long run?

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(A)

Number of Firms	Price
Increase	Increase

(B)

Number of Firms	Price
Decrease	Increase

(C) ✓

Number of Firms	Price
Increase	Decrease

(D)

Number of Firms	Price
Decrease	Decrease

(E)

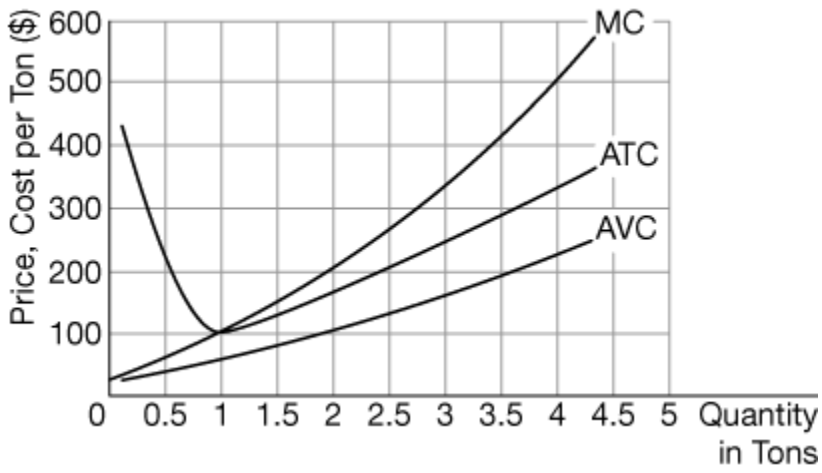
Number of Firms	Price
No change	Decrease

Answer C

Correct. The positive economic profits will attract new firms to enter the market. The entry of new firms increases the number of firms, which increases supply and decreases the price. The price will continue to fall until it is equal to minimum average total cost for each firm and economic profits are eliminated.

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78.



The graph above shows the marginal cost (MC), average total cost (ATC), and average variable cost (AVC) curves for a firm. If the market price is $\$100$ per ton, the firm will maximize profits or minimize losses by doing which of the following?

(A) Producing 0 tons and shutting down in the short run

(B) Producing 0 tons and exiting the market in the long run

(C) Producing 1 ton and earning zero economic profit ✓

(D) Producing 1.5 tons and earning zero economic profit

(E) Producing 2 tons and earning zero economic profit

Answer C

Correct. To maximize profits, firms should produce the quantity at which marginal revenue equals marginal costs $(\mathrm{MR} = \mathrm{MC})$. This is achieved when the firm produces 1 ton. At that quantity, price also equals average total cost $(\mathrm{P} = \mathrm{ATC} = \$100)$, so the firm will break even and earn zero economic profit.

79. A perfectly competitive firm finds that for the last shirt that it produces and sells, the marginal revenue is $\$20$ and the marginal cost is $\$22$. The firm should

(A) charge $\$22$ to maximize profit

(B) produce more shirts to increase profit

(C) produce fewer shirts to increase profit ✓

(D) hire more workers to increase production

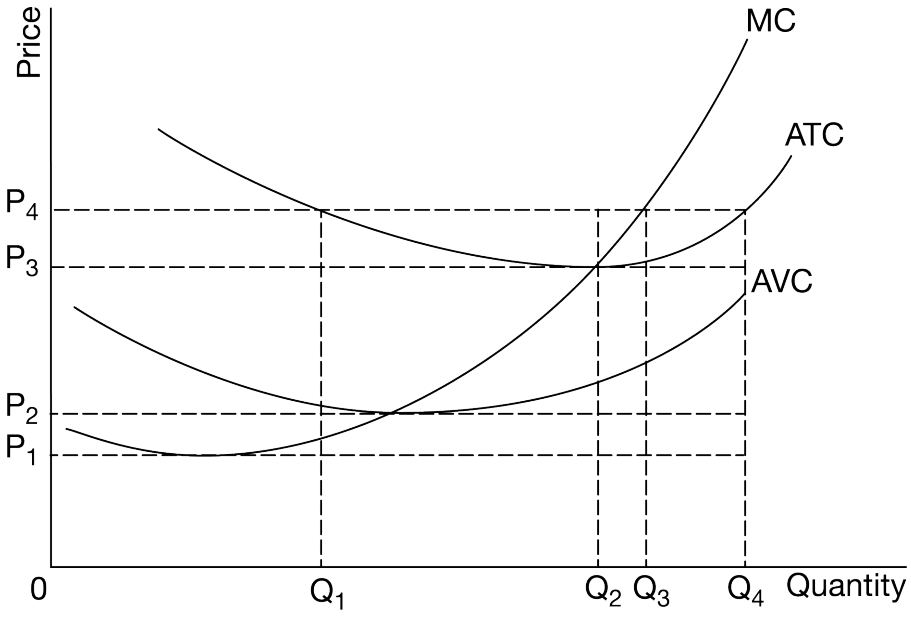
(E) shut down if the average variable cost is less than $\$20$

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Answer C

Correct. To maximize profits, firms should produce the quantity at which marginal revenue equals marginal cost $(\text{MR} = \text{MC})$. When $\text{MR} < \text{MC}$, it costs more to produce an additional shirt. Therefore, the firm should produce fewer shirts to increase profit.

The graph shows a firm in a perfectly competitive industry. Assume all the firms in the industry have identical costs.



80. Given the price P_4, what is a firm's profit-maximizing quantity of output?

(A) Q_1
(B) Q_2
(C) Q_3 ✓
(D) Q_4
(E) 0

Answer C

Correct. At Q_3 profit is maximized because price (marginal revenue) is equal to marginal cost. Producing more output beyond that will reduce total profit.

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81. Firms will have no incentive to exit or enter this market if the price in this market is

(A) above P_4
(B) below P_2
(C) between P_2 and P_3
(D) equal to marginal cost
(E) equal to average total cost ✓

Answer E

Correct. When the price is equal to the average total cost, the firm is earning zero economic profit and there is no motivation for the firm to exit the market or for new firms to enter the market.

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82. Suppose that a firm is producing the profit-maximizing output under conditions of diminishing returns. Its output price is $25, and its marginal cost of production at its current output level is $25. Based on this information, it can be concluded that this firm must

(A) be a single-price monopoly
(B) be a perfectly competitive firm ✓
(C) decrease its output to maximize profit it it is a monopoly
(D) increase its output to maximize profit if it is a monopoly
(E) exit the industry if it is a perfectly competitive firm

83. In a comparison of a profit-maximizing perfectly competitive firm's short-run equilibrium to its long-run equilibrium, which of the following is true?

(A) Price must equal marginal cost in the long run, but not necessarily in the short run.
(B) Economic profit must be positive in the long run, but not necessarily in the short run.
(C) The firm can set price in the short run, but not necessarily in the long run.
(D) The firm must produce at minimum average total cost in the short run, but not necessarily in the long run.
(E) Price equals average total cost in the long run, but not necessarily in the short run. ✓

Answer E

Correct. Perfectly competitive firms must earn zero profit in the long run, meaning price must equal marginal cost and average total cost. Profit can be positive, negative, or zero in the short run.
